The Universal Vacuum of the Fourth-Order Scalar Field: Metric Orbits, Fock Sectors, and the Krein Boundary

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Abstract

For the free fourth-order scalar field $(\Box+m_1^2)(\Box+m_2^2)\phi = 0$ we separate three structures that are commonly conflated: the geometry of positive metric operators, the geometry of vacuum functionals, and the geometry of the dynamics. Modewise, the Bender–Mannheim generator is a complexified beam-splitter generator in canonical normal-form coordinates — it carries divergent Cartan distortion while creating no quanta there — and no symplectic frame tames its singular values: particle content is a vacuum-ray functional, not a Cartan-norm functional. The entire family of admissible positive metrics fixes the physical vacuum ray and induces one and the same selected complex covariance — the *universal selected quasifree functional* within the stated real-form class; it carries exactly 1/3 of a quantum per ultraviolet mode pair relative to the two-field Fock vacuum, and, on the common auxiliary canonical algebra, its representation is disjoint from the two-field Fock representation in every spatial dimension — an obstruction no choice of metric can cure. The ultraviolet Fock-sector hierarchy of selected vacua terminates: no ultraviolet invariant for $d \leq 3$, the invariant $m_1^2 + m_2^2$ for $4 \leq d < 8$, the unordered mass pair for $d \geq 8$; global equivalence involving the doubly massless vacuum additionally requires an infrared analysis, which we identify but do not perform. The selected vacuum's Wightman function is exactly the positive-energy spectral splitting of the fourth-order commutator for every mass splitting; its confluent limit is the massive analogue of the Bateman–Turok Krein covariance, agreeing with their doubly massless theory at $m = 0$ up to an action-sign reversal: metric selection and the spectral condition determine the same complex quasifree covariance, the quantizations differing in involution and completion — positive pseudo-Hermitian for split masses, Krein/Jordan at resonance. Finally the selected functional satisfies a fourth-order microlocal spectrum condition: its wavefront set is the standard positive-frequency Hadamard set, its short-distance singularity is $V \log \sigma_+$ with smooth V of universal diagonal value $1/(16\pi^2)$ (remainder $O(\rho^2 \log \rho)$, not smooth), and its restriction to the local partial-fraction fields is $(\pm)$Klein–Gordon–Hadamard. All identities are machine-verified.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is an experiment in whether a non-specialist human, using

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AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

1 Introduction

The Pais–Uhlenbeck (PU) oscillator and its field-theoretic parent, the free fourth-order scalar

$$(\Box + m_1^2)(\Box + m_2^2)\,\phi = 0, \qquad m_1 > m_2 \geq 0, \tag{1}$$

admit two apparently different resolutions of the Ostrogradsky ghost problem. In the PT-symmetric approach of Bender and Mannheim [1, 2], each mode carries a positive metric operator $\eta = e^{-Q}$ that renders the dynamics unitary on a Hilbert space; in the Krein-space approach, recently developed for the pure fourth-order theory $\Box^2\phi = 0$ by Bateman and Turok [3], the state space is indefinite and probabilities are controlled by a hidden ghost-parity symmetry. Two companion papers [4, 5] analyzed the modewise PT construction as finite-dimensional symplectic geometry: the positive metric is unique up to a stabilizer orbit, is selected variationally by minimum Cartan distortion, and the modewise theory is exactly solvable in all the respects used below. The minimum-distortion selection is itself a Cartan-projection theorem, $F(S) = 2\|\mu(S)\|^2$ with $\mu(S_+) = (r, r)$ on the C_2 Weyl-chamber wall — but not a routine nearest-point (Mostow/Kempf–Ness) application: the stabilizer $SO(2,\mathbb{C})^2$ is not compatible with the Cartan involution ($H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$), and the proof passes through the θ-compatible hull $SL(2,\mathbb{C}) \times SL(2,\mathbb{C})$ [5].

This paper carries that analysis to the field theory, where three different geometries are in play and must not be conflated:

$$\text{metric geometry} \;\neq\; \text{vacuum geometry} \;\neq\; \text{dynamical geometry}. \tag{2}$$

The metric geometry is the symmetric-space geometry of positive operators, G/K for $G = \mathrm{Sp}(2n,\mathbb{C})$, where the Cartan projection μ measures nonunitarity [5]. The vacuum geometry is the geometry of Gaussian vacuum rays, which is governed by a different quotient: the stabilizer $P_\Omega = \{g : \hat{g}\,\Omega \in \mathbb{C}^\times\Omega\}$ of the vacuum line is the Siegel parabolic of the vacuum Lagrangian in the oscillator representation, and vacuum rays live in G/P_Ω. The vacuum-ray map $q_\Omega : G \to G/P_\Omega$ does not factor through $q_K : G \to G/K$ — no G-equivariant F with $q_\Omega = F \circ q_K$ exists, because the compact $K \cong \mathrm{Sp}(n)$ does not preserve the vacuum ray (only $K \cap P_\Omega \cong U(n)$ does), so q_Ω is not constant on K-cosets; the correct geometry is the correspondence $G/K \leftarrow G \to G/P_\Omega$ of two inequivalent quotient maps. The dynamical geometry is the spectral structure of the Hamiltonian, which degenerates (Jordan blocks) exactly at the resonant point $m_1 = m_2$ [4]. Our results, in brief:

1. *Cartan–parabolic decoupling* (Section 2). In canonical normal-form coordinates the Bender–Mannheim generator is a complexified beam-splitter generator, $Q' = r\,(a_1 a_2^\dagger + a_1^\dagger a_2)$: it lies in the complexified number-preserving (Levi) algebra of P_Ω — $e^{-Q'/2}$ with real Q' is *not* a unitary $SU(2)$ beam splitter — fixes the canonical vacuum exactly, and yet has Cartan projection (r, r) with $r = \log\frac{\omega_1+\omega_2}{\omega_1-\omega_2} \to \infty$ in the ultraviolet. Moreover no symplectic frame tames the singular values. Hence particle content is not a function of the Cartan projection, and no coercive lower bound vacuum excitation $\geq f(\|\mu\|)$ with

$f(r) \to \infty$ can hold: large Cartan distortion does not force particle content. Geometrically, the ultraviolet limit is escape to infinity along the C_2 chamber wall *within a single vacuum-ray fibre* of q_Ω.

2. *Orbit constancy and a universal obstruction* (Section 3). The physical annihilator covectors are eigenvectors of the stabilizer generators, so the entire admissible metric family — the Gaussian orbit and all $W = f(N_1, N_2) > 0$ — fixes the physical vacuum ray. The vacuum functional is therefore metric-independent. Its occupation relative to the two-field Fock vacuum is exactly

$$\langle N \rangle(k) = \frac{\omega_2(\omega_1^2 + \omega_2^2)}{2\omega_1(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)} \xrightarrow[k\to\infty]{} \frac{1}{3}, \tag{3}$$

a universal constant; the total relative quanta scale as $\Theta(V\Lambda^d)$ and the representations are disjoint in every spatial dimension. No choice of metric can cure this.

3. *Terminating ultraviolet sector hierarchy* (Section 4). Selected vacua of different mass pairs satisfy, in common field variables,

$$1 - f(k) = \frac{(\Sigma - \bar{\Sigma})^2}{12\,k^4} + O(k^{-6}), \qquad \Sigma = m_1^2 + m_2^2. \tag{4}$$

At matched Σ the k^{-6} coefficient also vanishes, and the first remaining obstruction is $\frac{29}{576}\,(\delta^2 - \bar{\delta}^2)^2 k^{-8}$, $\delta = \frac{1}{2}(m_1^2 - m_2^2)$; the invariants determine the mass pair only up to interchange, $(m_1^2, m_2^2) \sim (m_2^2, m_1^2)$. Hence the *ultraviolet* hierarchy: no UV invariant for $d \leq 3$; Σ for $4 \leq d < 8$; the unordered pair (Σ, Π) for $d \geq 8$; and the classification terminates. Global (GNS) equivalence involving the doubly massless vacuum requires a separate infrared analysis: against the $\Box^2$ anchor the relative occupation diverges as $(m_1{+}m_2)m_1m_2/(6k^3)$ in the infrared, and $\int_0 k^{d-1} N_{\mathrm{rel}}\,dk$ diverges precisely for $d \leq 3$.

4. *The spectral bridge* (Section 5). In the action convention of (7), the selected vacuum's Wightman function is exactly the positive-energy spectral splitting of the fourth-order commutator,

$$\widetilde{W}^+_{12}(p) = \varepsilon\,\theta(p^0)\,\frac{\delta(p^2 - m_1^2) - \delta(p^2 - m_2^2)}{m_1^2 - m_2^2}, \qquad \widetilde{W}^+_{mm}(p) = -\varepsilon\,\theta(p^0)\,\delta'(p^2 - m^2), \tag{5}$$

for all $\Delta = m_1^2 - m_2^2 \geq 0$. The confluent case is a *massive Bateman–Turok-type* Krein/Jordan covariance; at the doubly massless point $m = 0$, and after matching the action sign and infrared extension, it agrees with the covariance of [3] — in our $\varepsilon = +1$ convention it is its action-sign reversal. Metric selection and the spectral condition determine the same complex quasifree covariance; the quantizations differ in involution and completion — positive η-inner product where it exists ($\Delta > 0$), Krein space with ghost parity where it must ($\Delta = 0$). The covariance is analytic across the resonant boundary while the metric and diagonalizability fail there.

5. *Fourth-order microlocal spectrum condition* (Section 6). W^+_{12} is a bisolution with the correct fourth-order commutator, its wavefront set is the standard positive-frequency Hadamard set $\mathcal{C}^+$, and its short-distance singularity is logarithmic with universal diagonal

coefficient: the $1/\rho^2$ singularity of the Klein–Gordon factors is mass-independent and cancels, leaving the Hadamard form $W^+_{12} = V_{12}\log\sigma_+ + H_{12}$ with V_{12}, H_{12} smooth and V_{12} of universal coincidence value; in flat coordinates

$$W^+_{12}(x) = \frac{1}{8\pi^2}\log\rho + \frac{\Sigma}{64\pi^2}\,\rho^2\log\rho + \cdots + C^\infty, \tag{6}$$

i.e. leading coefficient $1/(16\pi^2)$ for $\log\rho^2$ with a remainder that is $O(\rho^2\log\rho)$ — milder than the leading logarithm but *not* smooth for nonzero masses. Its restrictions to the local partial-fraction fields are exactly $(\pm)$Klein–Gordon Hadamard functions. The fourth-order parametrix $H_P = (H_{m_1} - H_{m_2})/(m_1^2 - m_2^2)$ provides the local microlocal input for developing Wick powers and perturbative time-ordered products.

Together these give the corrected physical picture: *the universal selected vacuum is a fourth-order Hadamard quasifree functional; its apparently infinite particle content relative to the split two-field vacuum reflects inequivalent complex structures, not pathological local ultraviolet singularities* — and the Krein formulation is not the remnant of a divergent positive theory but a different completion of the same, analytically continued covariance.

What "universal" means. Throughout, the universal object is the *selected complex quasifree covariance*: it is constant over the entire admissible metric orbit (Theorem 3.2) and analytic in the masses across the resonant boundary. It is *not* claimed to be positive under one common involution across that boundary; its global Fock realization depends on dimension and infrared behaviour (Section 4); "particle number" is defined only relative to a chosen reference complex structure; and the massive confluent theory is the massive analogue of — not identical to — the doubly massless Bateman–Turok theory.

Verification. Every identity below — the beam-splitter identity, the occupation (3), the eigenvector/constancy lemma, the sector expansions with their exact coefficients, the Wightman functions and their confluent limits, the commutator normalizations, the logarithmic coefficient and the $\rho^2\log\rho$ remainder coefficient, and the infrared divergence of the relative occupation against the massless anchor — is machine-verified symbolically, with independent numerical checks. The artifact is the root of the `area9innovation/weyl-gravity` repository accompanying [4, 5] (scripts `symbolic/verify_paper3_audit.py` (P1–P10), `symbolic/verify_hadamard.py` (H1–H6), `symbolic/verify_paper3_referee.py` (R1–R5); SymPy 1.14.0, Python 3.14; the release tag `paper3-v1.3` fixes the commit).

Conventions. The theory is defined by the action

$$S[\phi] = \frac{\varepsilon}{2}\int \phi\,(\Box + m_1^2)(\Box + m_2^2)\,\phi, \qquad \varepsilon = +1 \text{ fixed throughout}, \tag{7}$$

whose canonical structure fixes the fourth-order commutator normalization $E_P'''(0) = +\varepsilon$; the equation of motion (1) alone does not fix the symplectic sign, and every signed object below (commutator, Wightman function, logarithmic coefficient, partial-fraction signature, comparison with [3]) carries ε. We further distinguish three layers: the *complex covariance* (a bilinear functional on the complex field algebra generated by the smeared field with no conjugation selected), the *involution* (a real form of that algebra), and the *positive/Krein* *completion* built on the involution; "same functional, different completion" always refers to this hierarchy. Modewise notation follows [4, 5]: for spatial momentum k, the mode is a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$, canonical variables $V = (x, y, p, q)$ related to the field variable by $z = iy$ (the PT rotation), normalized coordinates with $d_x d_y = \omega_1\omega_2$, rapidity $r(k) = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2} = \log \frac{(\omega_1+\omega_2)^2}{\Delta}$, selected diagonalizer $S_+ = e^{(r/2)B}$ and metric $\eta = e^{-Q}$, $Q = \alpha pq + \beta xy$. We write $\Delta = m_1^2 - m_2^2$, $\Sigma = m_1^2 + m_2^2$, $\Pi = m_1^2 m_2^2$. *Cartan convention*: $\mu(S)$ denotes the Gram–Cartan data $\mu(S) = \log \mathrm{spec}(S^\dagger S)$ restricted to the closed chamber, i.e. *twice* the standard singular-value projection $\mu_{\mathrm{std}}(S) = \log s(S)$; the singular values of $S_+ = e^{(r/2)B}$ are $e^{\pm r/2}$, so $\mu(S_+) = (r, r)$ while $\mu_{\mathrm{std}}(S_+) = (r/2, r/2)$. This is the convention synchronized with the distortion dictionary $F(S) = \| \log S^\dagger S\|_F^2 = 2\|\mu(S)\|^2$ of [5].

2 Metric geometry versus vacuum geometry

2.1 The beam-splitter identity

In the canonically normalized coordinates the mode annihilation operators of the auxiliary canonical vacuum are a'_j with $x' = (a'_1 + a'^{\dagger}_1)/\sqrt{2}$, etc. A one-line computation (different modes commute, so no ordering terms arise) gives the exact operator identity

$$Q' = r\,(x'y' + p'q') = r\,\big(a'_1 a'^{\dagger}_2 + a'^{\dagger}_1 a'_2\big). \tag{8}$$

Thus Q' lies in the complexified number-preserving beam-splitter algebra — the Levi algebra of the vacuum-line parabolic P_Ω; note that $e^{-Q'/2}$ with real Q' is *not* a unitary $SU(2)$ beam splitter, the generator being Hermitian rather than anti-Hermitian — with $[Q', N_{\mathrm{tot}}] = 0$ and $Q'\,\Omega_{\mathrm{can}} = 0$, so

$$e^{-Q'/2}\,\Omega_{\mathrm{can}} = \Omega_{\mathrm{can}} \qquad \text{exactly,} \tag{9}$$

even though the Cartan projection of the implemented symplectic map is $\mu(S_+) = (r, r)$ with $r(k) \sim 2\log k \to \infty$. Equivalently: Q' lies in the complexified number-preserving algebra, the Levi component of the parabolic-type stabilizer P_Ω of the vacuum line, on which the Cartan norm is unbounded.

2.2 No frame tames the map; the occupation is a state functional

One might hope that a frequency-adapted symplectic frame T_ω (the one whose canonical vacuum is the physical two-field vacuum) turns the divergent Cartan data into bounded physical squeezing. It cannot:

Lemma 2.1 (No frame tames the map)**.** *For every symplectic frame T (so that the Cartan data of $T^{-1}S_+T$ are defined in the same group), in the Gram convention of Section 1,*

$$\big\|\mu\big(T^{-1}S_+T\big)\big\|_\infty \;\geq\; \log \varrho\big((T^{-1}S_+T)^2\big) = \log \varrho(S_+^2) = r, \tag{10}$$

where ϱ is the spectral radius: for any matrix M, $\|\mu(M)\|_\infty = 2\log s_{\max}(M) \geq 2\log\varrho(M) = \log\varrho(M^2)$, and similarity preserves the spectrum of the positive S_+, which is $\{e^{\pm r/2}\}$. Hence $\|\mu(T_\omega^{-1}S_+T_\omega)\| \to \infty$ in the ultraviolet for every choice of frames T_ω. (Numerically the physical-frame Cartan data are in fact close to (r, r); the lemma is what the theorems below use.)

What is bounded is the *state*: the selected vacuum, computed exactly in [5] as the Gaussian ground state of the PT mode Hamiltonian, has occupation (3) relative to the two-field vacuum, saturating at 1/3. The two statements are compatible because particle content depends on the transformation only through the vacuum ray it produces — a G/P_Ω datum — while μ is a G/K datum, and the two are inequivalent quotients of G.

Theorem 2.2 (Cartan–parabolic decoupling)**.** *Let $G = \mathrm{Sp}(2n, \mathbb{C})$ act metaplectically on Fock space, and let $q_K : G \to G/K$ and $q_\Omega : G \to G/P_\Omega$, $q_\Omega(g) = [\hat{g}\,\Omega]$, be the two quotient maps. Then: (i) positive-metric geometry is a function of q_K (the Cartan projection); (ii) Gaussian vacuum rays are a function of q_Ω; (iii) q_Ω does not factor through q_K — there is no G-equivariant F with $q_\Omega = F \circ q_K$ — since $K \not\subseteq P_\Omega$ ($K \cap P_\Omega \cong U(n) \subsetneq K \cong \mathrm{Sp}(n)$), so q_Ω is not constant on K-cosets; (iv) the Cartan norm is unbounded on P_Ω (already on its Levi component), so fibres of q_Ω contain elements of arbitrarily large Cartan norm; consequently (v) particle content is not a function of the Cartan projection, and no coercive lower bound $\langle N \rangle \geq f(\|\mu\|)$ with $f(r) \to \infty$ holds in general. (We do* not *claim that no upper envelope $\langle N \rangle \leq f(\|\mu\|)$ exists; standard Bogoliubov estimates bound particle number above by functions of the singular values, and on bounded Cartan sets Gaussian occupation is continuous. The decoupling is one-sided: distortion does not* force *excitation.) The PU metric orbit realizes the extreme case: $q_\Omega(S_+H) = \{[\Omega_{\mathrm{PT}}]\}$ is a single vacuum ray (Theorem 3.2), while $q_K(S_+H)$ is unbounded, with $\mu(S_+) = (r, r)$ divergent in the ultraviolet yet identically vanishing canonical-frame excitation, and bounded physical excitation* (3) *despite Lemma 2.1.*

The proof of (iv) is the identity (8) together with $\mu(e^{(r/2)B}) = (r, r)$; (v) follows because the exhibited family has $\|\mu\| \to \infty$ with vanishing canonical-frame excitation, refuting any coercive lower bound. This corrects a natural but false expectation — used, e.g., to estimate Bogoliubov costs by $\sinh^2$ of Cartan coordinates — and it is the conceptual reason the theorems below must be formulated at the level of vacuum functionals.

Remark 2.3 (The geometric picture). In the Cartan-projection formulation of [5] (Section 1), $\mu(S_+) = (r, r)$ lies on the C_2 Weyl-chamber wall for every k, and the ultraviolet limit $r(k) \sim 2 \log k \to \infty$ is escape to infinity *along the wall* while remaining inside a single fibre of the vacuum-ray map q_Ω. This is the sharpest form of Cartan–parabolic decoupling: the metric representative travels unboundedly in G/K along a fixed direction while the vacuum ray never moves.

3 Orbit constancy and the universal Fock obstruction

Lemma 3.1 (Eigenvector lemma)**.** *Let ℓ_j be the physical annihilator covectors (the two-field vacuum modes) and X_j the stabilizer generators of [5]. Then*

$$\ell_j^\top X_j = -i\,\ell_j^\top, \qquad \textit{hence} \qquad \ell_j^\top C(\theta_1, \theta_2)^{-1} = e^{i\theta_j}\,\ell_j^\top \quad (\theta_j \in \mathbb{C}). \tag{11}$$

Theorem 3.2 (Metric-independence of the vacuum)**.** *Call a positive metric* admissible *if it belongs to the classification $\eta' = \rho^\dagger W \rho$ of [4], with $W = f(N_1, N_2) > 0$ densely defined with the polynomial CCR algebra of the mode variables in its domain, and Ω an eigenvector of W (automatic for $W = f(N_1, N_2)$: $W\Omega = w_0\Omega$, $w_0 = f(0,0) > 0$). Then the selected physical vacuum ray, and the associated quasifree functional on the polynomial CCR algebra, is the same for every admissible positive metric: (i) on the Gaussian orbit η_C, $C \in SO(2, \mathbb{C})^2$, the* *annihilation conditions defining the vacuum are unchanged by Lemma 3.1; (ii) the metaplectic implementers of the stabilizer are of the form* $\exp[\theta_1(N_1+\frac{1}{2})+\theta_2(N_2+\frac{1}{2})]$ *and act on the vacuum ray by scalars; (iii) for the full family,* $W^{\pm 1/2}$ *also fixes the vacuum ray. Hence* $[\Omega_\eta] = [\Omega_{\eta_+}]$ *for every admissible* η*.*

Ray constancy implies functional constancy. Constancy of the ray alone does not suffice, because a metric change alters both the physical inner product and the representatives of observables; the functional is unchanged because the two changes cancel. With $\rho' = W^{1/2}\rho$ and the functional defined by pullback on the common polynomial CCR domain,

$$\omega_{\rho'}(A) = \frac{\langle \Omega,\, W^{1/2}\rho\, A\, \rho^{-1} W^{-1/2}\, \Omega\rangle}{\langle \Omega, \Omega\rangle} = w_0^{1/2}\, w_0^{-1/2}\, \frac{\langle \Omega,\, \rho A \rho^{-1}\, \Omega\rangle}{\langle \Omega, \Omega\rangle} = \omega_\rho(A), \tag{12}$$

using $W^{\pm 1/2}\Omega = w_0^{\pm 1/2}\Omega$ on both sides of the matrix element. The identity holds for every A in the polynomial CCR algebra; unbounded W enters only through its action on Ω and on the (analytic-vector) domain. □

Metric changes do alter the physical inner product and the representatives of observables individually; the displayed cancellation is what makes any quantity determined by the fixed vector relative to the fixed two-field reference — in particular the occupation (3) and the fidelity below — metric-independent.

Theorem 3.3 (Universal Fock obstruction)**.** *Transport both constructions to the common auxiliary real CCR algebra of the canonical variables* $V = (x, y, p, q)$ *per mode, with its standard involution — on this algebra both the two-field Fock vacuum and the selected vacuum are positive Gaussian states (the selected vacuum is a normalizable real Gaussian [5]); the physical involution differs by the PT rotation* $z = iy$*, which is genuinely complex, and this is why positivity of the selected functional in* physical *variables is available only in the* η*-inner-product sense. Then, for every admissible positive metric, in a volume* V *with cutoff* Λ*: the relative quanta obey* $N(\Lambda) \sim C_d V \Lambda^d$ *with* $C_d > 0$*; the vacuum fidelity obeys* $|\langle \Omega_{2\text{-}field} | \Omega_{\mathrm{PU}}\rangle|^2 \le e^{-cV\Lambda^d}$ *(per-mode fidelity* $\to \sqrt{3}/2$ *[5]); and in the thermodynamic/continuum limit the corresponding positive representations of the common auxiliary algebra are disjoint, in every spatial dimension* $d \ge 1$*. Metric ambiguity cannot cure the inequivalence.*

Proof. Combine Theorem 3.2 with the exact modewise data (3) and the von Neumann infinite-tensor-product criterion, as in [5]; disjointness is meant for the stated common auxiliary algebra, on which both states are positive under one involution. □

4 The ultraviolet sector hierarchy

Comparisons between selected vacua of different mass pairs are made in common physical variables ($\gamma = 1$ PU variables per mode; the normalization map contributes only $O(k^{-2})$). The exact modewise Gaussian of [5] gives, for pairs (m_1, m_2) and $(\bar{m}_1, \bar{m}_2)$, the fidelity expansion (4); at matched Σ the expansion continues

$$1 - f(k)\Big|_{\Sigma=\bar{\Sigma}} = \frac{29}{576}\,\frac{(\delta^2 - \bar{\delta}^2)^2}{k^8} + O(k^{-10}), \qquad \delta^2 - \bar{\delta}^2 = \bar{\Pi} - \Pi, \tag{13}$$

i.e. at matched Σ the k^{-6} coefficient also vanishes and the first remaining obstruction occurs at order k^{-8}.

Theorem 4.1 (Ultraviolet sector hierarchy)**.** *Let d be the spatial dimension, and hold the infrared behaviour of the compared vacua in a common regular class (in particular, strictly positive masses). The relative ultraviolet Bogoliubov transformation is Hilbert–Schmidt — diagnosed by $\int^{\Lambda} k^{d-1}(1-f(k))\,dk < \infty$, which for nearby ultraviolet Gaussians is proportional to the squared Bogoliubov coefficient — precisely according to:*

spatial dimension	*UV sector invariant of selected vacua*
$d \leq 3$	*none (no UV invariant)*
$4 \leq d < 8$	$\Sigma = m_1^2 + m_2^2$
$d \geq 8$	(Σ, Π)*, i.e. the unordered mass pair*

The hierarchy terminates: the two symmetric invariants determine the pair up to interchange $(m_1^2, m_2^2) \sim (m_2^2, m_1^2)$. At the critical dimensions the obstruction is logarithmic rather than power-law: at $d = 4$ (respectively $d = 8$) the divergent sum for $\Sigma \neq \bar{\Sigma}$ (respectively $\Pi \neq \bar{\Pi}$ at matched Σ) grows as $\log \Lambda$, so the critical dimensions belong to the labeled regimes as indicated. This theorem classifies ultraviolet (local) sectors; global equivalence additionally requires an infrared analysis, as follows.

Remark 4.2 (The $\Box^2$ anchor is an infrared question, left open)*.* The vacuum-overlap criterion is an ultraviolet diagnostic only: $1-f$ is bounded by 1, so its infrared integral always converges, whereas the global Shale criterion involves the relative occupation N_{rel}, which need not stay bounded. Against the doubly massless selected vacuum ($m_1 = m_2 = 0$; modewise Gaussian $A_0 = (2k, k^2; k^2, 2k^3)$), a massive selected vacuum has (machine-verified)

$$N_{\mathrm{rel}}(k) \;\sim\; \frac{(m_1+m_2)\,m_1 m_2}{6\,k^3} \qquad (k \to 0), \tag{14}$$

so $\int_0 k^{d-1} N_{\mathrm{rel}}(k)\,dk$ diverges precisely for $d \leq 3$: the dimensions with the most benign ultraviolet behaviour are exactly those in which the massless anchor carries the strongest potential infrared obstruction. Whether the $\Box^2$ vacuum lies in the same *global* Fock/GNS sector as the massive selected vacua therefore requires a separate infrared (Shale/Araki–Yamagami) analysis, which we do not perform here. For strictly positive masses the infrared is uniformly regular and the hierarchy of Theorem 4.1 stands as stated.

At the resonant boundary four kinds of regularity must be distinguished, and they are logically independent. *Distributional regularity*: the selected two-point functional is analytic in Δ across $\Delta = 0$ (modewise, $1 - f \sim \epsilon^4$ for $m_{1,2} = m \pm \epsilon$; the Wightman functions of Section 5 converge smoothly), and the occupation limits commute ($\langle N \rangle \equiv \frac{1}{3}$ at $\Delta = 0$ for every k). *Representation regularity at fixed positive mass*: the confluence $m_1, m_2 \to m > 0$ introduces no new infrared singularity, and for $d \leq 3$ the massive resonant vacuum lies in the same ultraviolet-dressed sector as every nondegenerate selected vacuum with the same infrared class (Theorem 4.1). *The doubly massless corner* $m = 0$ is a separate boundary governed by Remark 4.2 and is not settled by the ultraviolet hierarchy. *Dynamical and metric singularity*: as $\Delta \to 0$ the metric operator degenerates ($\alpha, \beta \to \infty$), the similarity transformation diverges ($S_+ \sim \Delta^{-1/2}$), and the dynamics becomes nondiagonalizable (Jordan blocks; no positive invariant form [4]). In summary,

$$\boxed{\text{resonance is not a vacuum singularity; it is a metric and dynamical singularity.}} \tag{15}$$

5 The spectral bridge to the Krein quantization

The physical Wightman function of the selected vacuum follows from the exactly solvable mode: with $\widetilde{V} = S_+ V$ evolving under the diagonalized Hamiltonian, $W(t) = S_+ e^{tA_0} \Gamma S_+^\top$ pulled back to the field variable $z = iy$ gives, exactly,

$$W_{zz}(t) = \frac{1}{\Delta}\left[\frac{e^{-i\omega_1 t}}{2\omega_1} - \frac{e^{-i\omega_2 t}}{2\omega_2}\right], \qquad W_{zz}(t)\Big|_{\Delta\to 0} = -\frac{(1+i\omega t)\, e^{-i\omega t}}{4\omega^3}. \tag{16}$$

Both expressions contain only positive frequencies; the first is the momentum-space spectral function (5), the second its confluent (δ') limit. The commutator parts reproduce the classical fourth-order Green-function difference $E_P = [\sin\omega_2 t/\omega_2 - \sin\omega_1 t/\omega_1]/\Delta$ with the normalization $E_P(0) = E_P'(0) = E_P''(0) = 0$, $E_P'''(0) = 1$, and confluent limit $(\sin\omega t - \omega t\cos\omega t)/(2\omega^3)$.

Theorem 5.1 (Spectral bridge)**.** *Fix the action sign and Fourier convention of* (7) *(*$\varepsilon = +1$*,* $E_P'''(0) = +1$*). Three statements, in decreasing generality:*

1. Split masses. *For* $m_1 \neq m_2$*, the selected PU covariance is the positive-energy spectral splitting of the fourth-order commutator,* (5)*: metric selection and the spectral condition determine the same complex quasifree covariance.*

2. General resonance. *For* $m_1, m_2 \to m$ *(any* $m \geq 0$*), the confluent limit is*

$$\widetilde{W}^+_{mm}(p) = -\varepsilon\, \theta(p^0)\, \delta'(p^2 - m^2) = \varepsilon\, \partial_{m^2}\left[\theta(p^0)\delta(p^2 - m^2)\right], \tag{17}$$

   a massive Bateman–Turok-type *Krein/Jordan covariance, realized by a Jordan mode doublet with null pairing* $[a_1, a_2^\dagger] = [a_2, a_1^\dagger] \neq 0$*. This massive family is not the theory treated in [3].*

3. Doubly massless point. *At* $m = 0$*, and after matching the infrared extension, the confluent covariance agrees with the Bateman–Turok covariance* $\widetilde{W}_{\mathrm{BT}}(p) = \theta(p^0)\delta_1'(p^2)$*,* $\delta_1'(p^2) = -\partial_{m^2}\delta(p^2 - m^2)|_{m^2=0}$ *[3], up to the overall action-sign reversal* $\varepsilon \to -\varepsilon$*: in our* $\varepsilon = +1$ *convention the two covariances are negatives of each other, and they coincide in the perfect-square convention* $\varepsilon = -1$*.*

For $\Delta > 0$ *the covariance is realized through a positive pseudo-Hermitian completion (*η*-inner product); at* $\Delta = 0$ *it is realized through the resonant Krein/Jordan completion with ghost parity. Equality throughout concerns the complex quasifree covariance (Conventions); we do not assert positivity with respect to a common involution.*

Proof. Statement 1 is the computation (16) (machine-verified, including the commutator normalization $E_P'''(0) = +\varepsilon$, which is what ties the covariance to the action sign). Statement 2 is the machine-verified confluent limit of (16): the divided difference converges to $+\partial_{m^2} W_m^+ = -(1 + i\omega t)e^{-i\omega t}/(4\omega^3)$ per mode. Statement 3 is then the identity $\delta_1'(p^2) = -\partial_{m^2}\delta(p^2 - m^2)|_0$: the Bateman–Turok mode function is the negative of ours, which is exactly the sign of their perfect-square action. Quasifree functionals are determined by their two-point distributions, so these mode-level identities identify the covariances as stated. □

Remark 5.2. (i) The theorem upgrades the resonant-boundary picture: the Krein formulation is not the residue of a positive theory destroyed by an ultraviolet particle catastrophe (Section 2 shows no such catastrophe exists); it is a different completion of the *same*, analytically continued covariance, adapted to Jordan dynamics. (ii) At $\Delta > 0$ the completion ambiguity is invisible in expectation values but material for which operators are Hermitian. At $\Delta = 0$ no nondegenerate positive-definite invariant metric exists (the Jordan obstruction of [4]), and the Krein construction is the canonical nondegenerate spectral completion considered here; singular or intrinsically Jordan PT-symmetric boundary realizations in the sense of Bender and Mannheim [2] are distinct constructions not classified by this theorem. (iii) What remains genuinely open at the resonant point is the treatment of the generalized (Jordan) excitations, where the two completions differ by construction; ghost parity governs precisely that sector [3].

6 Hadamard structure of the selected vacuum

Let W_m^+ denote the Minkowski Klein–Gordon positive-frequency two-point function and define the divided difference and its confluent limit

$$W_{12}^+ = \frac{W_{m_1}^+ - W_{m_2}^+}{m_1^2 - m_2^2}, \qquad W_{mm}^+ = +\,\partial_{m^2} W_m^+ \tag{18}$$

(the confluent limit of a divided difference is the +derivative; machine-verified against (16)), which by Theorem 5.1 are the selected two-point functions in the $\varepsilon = +1$ convention.

Theorem 6.1 (Fourth-order Hadamard property)**.** *Let $P = (\Box + m_1^2)(\Box + m_2^2)$, $d = 3$ spatial dimensions. Then:*

1. *$P_x W_{12}^+ = P_y W_{12}^+ = 0$, and $W_{12}^+(x, y) - W_{12}^+(y, x) = iE_P(x, y)$ with $E_P = (E_{m_1} - E_{m_2})/\Delta$ the fourth-order commutator function;*

2. *$\mathrm{WF}(W_{12}^+) = \mathcal{C}^+$, the standard positive-frequency Hadamard wavefront set; the multiplicity of the characteristic polynomial changes the order of the light-cone singularity, not its wavefront directions;*

3. *the short-distance singularity is logarithmic, in the Hadamard form*

$$W_{12}^+(x, y) = V_{12}(x, y)\, \log \sigma_+(x, y) + H_{12}(x, y), \tag{19}$$

with V_{12} and H_{12} smooth and universal (mass-independent) nonzero coincidence value $V_{12}(x, x) = \frac{1}{16\pi^2}$ (in the $\log \rho^2$ normalization; $\frac{1}{8\pi^2}$ for $\log \rho$): with $W_m^+(x) = \frac{1}{4\pi^2} \frac{mK_1(m\rho)}{\rho}$, $\rho^2 = -x^2 + i0x^0$, the mass-independent $\frac{1}{4\pi^2 \rho^2}$ term cancels in the divided difference and, in flat coordinates,

$$W_{12}^+(x) = \frac{1}{8\pi^2} \log \rho + \frac{\Sigma}{64\pi^2}\, \rho^2 \log \rho + \cdots + C^\infty, \tag{20}$$

where the displayed subleading term (coefficient machine-verified) is milder than the leading logarithm but not *smooth for nonzero masses: the universal statement concerns the coincidence value of the smooth logarithmic coefficient, not constancy of the coefficient modulo smooth terms. The confluent case has the identical leading coefficient;*

4. *in the doubly massless case $P = \Box^2$ the infrared extension of $\theta(p^0)\delta'(p^2)$ requires a scale; different admissible scales change W by a smooth constant, affecting neither the wavefront set nor the local Hadamard property — the bridge of Theorem 5.1 presumes matched extension conventions;*

5. *any two fourth-order spectral Hadamard functionals with the same commutator differ by a smooth bisolution; the fourth-order parametrix $H_P = (H_{m_1} - H_{m_2})/\Delta$ (confluently $\partial_{m^2} H_m$), whose full smooth coefficient multiplies $\log \sigma_+$ as in (3), has $W_{12}^+ - H_P$ smooth, and provides the local microlocal input for developing Wick powers $:\phi^n:_{H_P}$ and perturbative time-ordered products. (The Hollands–Wald and Brunetti–Fredenhagen constructions are formulated for hyperbolic operators and positive Hadamard states; their adaptation to a fourth-order operator with repeated characteristics at resonance, an indefinite quasifree functional, and a nonstandard involution is a genuine further step, not claimed here.)*

Proof sketch. (1) is verified exactly modewise (both branches and the confluent Jordan mode), with the Green normalization $E_P'''(0) = 1$. For (2), Fourier support in the closed forward cone alone does not give the position-space wavefront relation; the inclusion WF $\subseteq \mathcal{C}^+$ follows because divided differences and ∂_{m^2} of the smooth family $m^2 \mapsto W_m^+$ preserve the Hadamard wavefront bound of the Klein–Gordon factors. For the reverse inclusion, use the boundary-value form of $\log \sigma_+$ (equivalently the explicit Hadamard expansion of (3)): the mass-independent $1/\sigma$ singularities cancel, which *lowers the singular order* but removes no covector directions, because the smooth coefficient V_{12} has nonzero diagonal value $1/(16\pi^2)$; hence no component of $\mathcal{C}^+$ disappears, and the result is a *fourth-order microlocal spectrum condition* (we avoid "Hadamard state", which normally presupposes positivity relative to a fixed involution). (3) is the Bessel expansion $mK_1(m\rho)/\rho = \rho^{-2} + \frac{m^2}{2}\log(m\rho/2) + \ldots$: the ρ^{-2} terms cancel; the $\log \rho$ coefficient of the divided difference is $\left[\frac{m_1^2}{2} - \frac{m_2^2}{2}\right]/\Delta \cdot \frac{1}{4\pi^2} = \frac{1}{8\pi^2}$ and the $\rho^2 \log \rho$ coefficient is $\Sigma/(64\pi^2)$ (both machine-verified, as is the confluent case). (4) is the observation that the scale enters only through $\log(\mu\rho)$, so a change of scale adds $\log(\mu/\bar{\mu})/(8\pi^2)$, a constant. (5) follows from (1)–(3) by the usual difference argument: the difference of two such functionals is a bisolution with vanishing antisymmetric part and empty wavefront set. □

Proposition 6.2 ((±)Hadamard split structure)**.** *Let $\phi_1 = (\Box + m_2^2)\phi/(m_2^2 - m_1^2)$ and $\phi_2 = (\Box + m_1^2)\phi/(m_1^2 - m_2^2)$ be the local partial-fraction fields. In the selected vacuum, exactly,*

$$W_{\phi_1\phi_1} = +\frac{W_{m_1}^+}{\Delta}, \qquad W_{\phi_2\phi_2} = -\frac{W_{m_2}^+}{\Delta}, \qquad W_{\phi_1\phi_2} = 0. \tag{21}$$

The ghost branch carries a Klein–Gordon–Hadamard *distribution with negative Krein signature* — not a positive KG state under the ordinary KG $*$-structure: the Krein signature is intrinsic to the selected functional, while its local singularity structure is standard on both branches. This makes the summary statement precise:

> *The universal selected vacuum is a fourth-order Hadamard quasifree functional. Its apparent infinite particle content relative to the split two-field vacuum reflects inequivalent complex structures — the positive-positive versus positive-negative pairing of the two branches — not pathological local ultraviolet singularities.*

The logarithmic singularity is the expected one for a dimension-zero field in four spacetime dimensions and is milder than the Klein–Gordon $1/\sigma$; local Wick composites relative to H_P are well-defined at the level of the subtraction, and derivative observables (in particular the stress tensor) require the usual local subtractions relative to H_P, with no evident ultraviolet obstruction. A systematic treatment along the lines of [10, 11, 12] for the fourth-order operator — which would have to accommodate repeated characteristics at resonance, indefiniteness, and the nonstandard involution — now has its local microlocal input in place.

7 Discussion

What the three geometries settle. The separation (2) resolves several tensions in the literature at once. The divergent rapidity $r(k) \sim 2\log k$ is real, but it is metric geometry: it makes the phase-space (observable) metric a Sobolev pair $H^1 \oplus H^{-1}$ [5] and the Dyson operator unboundedly non-unitary in every frame — yet it creates no quanta, because it acts inside the vacuum-line stabilizer. The physical inequivalence to the naive two-field Fock space is vacuum geometry: an $O(1)$-per-mode, metric-independent statement (Theorems 3.2, 3.3). The resonant $\Delta \to 0$ singularity is dynamical geometry: Jordan blocks and the death of the positive metric, with the vacuum analytic throughout (Theorem 4.1 ff.). Conflating any two of these produces false conclusions — e.g. ultraviolet particle catastrophes (Λ^{d+2} scalings) or mass-splitting superselection sectors, both of which are refuted by the exact computations here.

Relation to Bender–Mannheim and Bateman–Turok. Theorem 5.1 shows the two programs select the same complex covariance, from opposite directions: positivity of the inner product modewise (Bender–Mannheim), spectral positivity of the energy covariantly (Bateman–Turok). The difference is the completion, and the difference matters exactly where the completions are forced apart: at the resonant point, where positivity fails (the Jordan no-go of [4]) and ghost parity takes over the excitation sector. A sharp remaining question is the representation-theoretic status of the generalized Jordan excitations at $\Delta = 0$: the vacuum sector is common, the one-particle completions are not, and the interacting perfect-square theory of [3] lives precisely there.

The division of labor should be stated plainly. The Krein quantization itself, the hidden ghost-parity symmetry, the two-field $O(1,1)$ embedding, and the positivity of transition probabilities on an indefinite state space are constructions of Bateman and Turok [3]; none of them is claimed here. What this paper contributes is the identification of the common *free complex covariance* underlying both programs and the precise determination of which additional data — involution, observable algebra, and completion — distinguish them. Two consequent cautions. First, a common free complex covariance does not imply an identical interacting quantum theory: the companion interaction paper [7] shows the two completions *separate* under interaction — the canonical positive metric is obstructed by on-shell branch-changing conversion, while the Krein/ghost-parity structure survives the same interaction exactly. Second, the "Krein boundary" of the title is a statement about the free covariance: the confluent covariance naturally admits an indefinite/Krein completion; it does not by itself determine the interacting Bateman–Turok theory, whose $O(1,1)$, null-state, and ghost-symmetry structure is additional data beyond the covariance.

Toward quadratic gravity. The scalar results transfer directly to genuine PU blocks: the two transverse-traceless helicity blocks of scalar-free Einstein–Weyl gravity, and scalaron blocks when present. They do *not* transfer independently to every spin-two polarization: after diffeomorphism reduction the physical phase space stratifies as $\Gamma_{\text{phys}} \cong 2\Gamma_{\text{PU}} \oplus 2\Gamma_{-} \oplus \Gamma_{-}$ — only helicity ± 2 survives as genuine PU pairs, the massive helicities $\pm 1, 0$ are unpaired second-order ghosts (their massless partners lie in the diffeomorphism orbit), and Lorentz covariance then forces all five massive polarizations into one joint real-form choice [6]. Where the transfer applies, the physical vacuum is spectral, satisfies the fourth-order microlocal spectrum condition, is metric-independent, and is not normal to the naive two-field (graviton plus ghost) Fock representation; PT-based unitarity claims should accordingly be phrased in the dressed representation, and interacting constructions must fix the representation before the perturbative expansion. We leave the interacting analysis, the stress-tensor subtractions relative to H_P, and curved backgrounds to future work.

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